NYC Mobility Analytics Platform

ITI – Data Visualization & Integration Track – Graduation Project

1. Background

You are tasked with building an end-to-end data engineering solution for a mobility analytics company that wants to analyze trip demand, operational patterns, and fare-related metrics across New York City.

Your goal is to transform raw trip records into a reliable data warehouse and provide meaningful business insights through an interactive Apache Superset dashboard.

2. Project Objectives

- Design and implement a dimensional data model.
- Build an ETL pipeline to ingest, clean, transform, and load the data.
- Identify and handle real-world data quality issues.
- Schedule and orchestrate dependent jobs with retries, logging, and failure handling.
- Implement historical backfills and safe reruns.
- Optimize analytical query performance.
- Build an interactive dashboard using Apache Superset.
- Present the complete solution and its business value.

3. Data Source

Dataset: NYC High-Volume For-Hire Vehicle (HVFHV) Trip Records.

Official download page: https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page

Use the official monthly Parquet files and the relevant data dictionary and taxi-zone lookup tables.

Requirements:

- Process approximately **10 GB of raw data or from 2017 till now**.
- Use HVFHV records only; do not substitute **Yellow Taxi or other datasets**.
- Select available monthly files up to August 2026, subject to the agreed dataset size.
- HVFHV records begin in February 2019; do not assume data is available from 2017.
- **Document** the selected files, date range, and total raw size.

Verify the availability and combined size of your selected files before starting.

4. Data Quality and ETL

Identify, measure, and handle actual data quality issues, such as:

- Missing values and invalid timestamps.
- Duplicate records.
- Negative or implausible trip durations and distances.
- Inconsistent fare-related values.
- Unknown or unmatched location identifiers.
- Schema differences between monthly files.

Document your transformation rules and justify decisions to correct, quarantine, or exclude records.

Your ETL pipeline must support reproducible processing, safe reruns, logging, and data-quality validation.

5. Dimensional Modeling

Design a dimensional model that supports the required business questions.

Define the grain, keys, relationships, measures, and handling of unknown dimension values. Include a dimensional-model diagram and explain your design decisions.

6. ETL Orchestration

Use an orchestration tool such as Apache Airflow to automate your pipeline.

Requirements:

- Schedule jobs to run automatically.
- Define dependencies between tasks.
- Implement retries, logging, and failure handling.
- Support safe reruns and historical backfills.
- Demonstrate a failed execution and explain how it is recovered.
- Document the schedule and the pipeline's execution flow.

7. Performance Optimization

Implement at least **two** meaningful optimization, such as partitioning, indexing, pre-aggregated tables, materialized views, or caching.

Explain why you chose it and provide measurements or reproducible evidence of its impact where possible.

8. SQL Analytics

Develop **at least 8 non-trivial SQL queries** that answer business questions.

Examples:

- How does trip demand vary by hour and day?
- Which pickup zones have the highest trip volumes?

For each query, provide the business question, SQL, results, and interpretation.

9. Apache Superset Dashboard

Build an interactive dashboard connected to your data warehouse.

Include relevant KPIs and visualizations, such as:

- Total trips and demand trends.
- Demand by hour, day, and location.

Add suitable filters, clear KPI definitions, and meaningful business insights.

Bonus: Customize the dashboard with advanced metrics, additional interactions, improved layouts, or other useful enhancements.

10. Documentation Deliverables

Submit a technical report containing:

- **Introduction:** Business context, objectives, and requirements.
- **Data sources:** Official URLs, selected files, date range, size, and limitations.
- **Architecture:** End-to-end architecture and technology choices.
- **Transformations:** Data quality issues, cleaning rules, and validation results.
- **Data warehouse model:** Fact tables, dimensions, relationships, and model diagram.
- **ETL and orchestration:** Job dependencies, scheduling, retries, and backfills.
- **Performance:** Optimization approach and supporting measurements.
- **SQL analytics:** Five or more queries, results, and business interpretation.
- **Dashboard:** Screenshots, KPIs, filters, and key findings.
- **Appendix:** SQL scripts, relevant code, setup instructions, and external references.

11. Final Presentation

Present and demonstrate the complete solution, covering:

1. Business problem and objectives.
2. Data source and technical architecture.
3. Dimensional model and data transformations.
4. ETL pipeline and orchestration.

5. Failure recovery and historical backfill.
6. Performance optimization.
7. SQL analytics and business findings.
8. Interactive Apache Superset dashboard.

12. Evaluation Criteria

Area	Weight
Data ingestion and storage	10%
Data quality and transformations	15%
Dimensional modeling	15%
ETL and orchestration	20%
SQL analytics	10%
Apache Superset dashboard	10%
Performance optimization and validation	10%
Documentation and presentation	10%
Total	**100%**

Final Expectation

Deliver a complete, reproducible, and documented data engineering platform that transforms raw data into a reliable analytical warehouse and provides meaningful business insights—not merely a collection of scripts or a dashboard.

13. Project Deadline

All interns must submit their completed projects and presentations, and demonstrate their ETL pipelines, by **5/11 "5th November 2026"**.